

BIMM-143: INTRODUCTION TO BIOINFORMATICS

The find-a-gene project assignment

<http://thegrantlab.org/bimm143>

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Overview:

The find-a-gene project is a required assignment for BIMM-143. You should prepare a written report in **PDF** format that has responses to each question labeled **[Q1] - [Q10]** below. You may wish to consult the scoring rubric at the end of this document and the example report provided online (note that the example report is from a previous quarter and the questions may differ).

The objective with this assignment is for you to demonstrate your grasp of database searching, sequence analysis, structure analysis and the R environment that we have covered in class.

Due Date:

Your responses to questions Q1-Q4 are due at 12pm on the **Monday of Week 5** (see the Assignments and Grading section of our website for details). Note that these first set of answers can be obtained very quickly (at best within 15 or 20 minutes), so if you don't succeed at first, just keep trying.

The complete assignment, including responses to all questions, is due at 12pm on the **Monday of Week 10.**

Submission instructions:

Your report formatted as a **PDF document** should be uploaded to **GradeScope**. Please make sure to include your UCSD email and PID number on the first page.

Be sure to include your UCSD email and PID number on the first page of your report.

Submit your preliminary report with answers to Q1-Q4 as soon as you can so we can determine if you have found a novel gene. Submit this preliminary report as one document with screen shots of the results inserted appropriately.

See the demonstration report linked to on the course website for an example of format. Note again that example questions may differ. I will indicate on GradeScope my decision (1pt indicating all is good, 0pts revisions required). You should proceed with subsequent questions only after we are sure you have found a novel gene (and thus be successful in the later stages of the project).

For the final report add your results for Q5-Q10 to the preliminary report and submit the final document containing your results for all questions.

Please do not send only Q5-Q10 answers as the final report.

Questions:

[Q1] Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as its function is known.

If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

Protein name: Retinol-binding protein 4 (RBP4)
Species: Homo sapiens
Accession number: P02753
Known Function: Retinol binding and retinol transmembrane transporter activity

[Q2] Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: TBLASTN (2.17.0+) search against amphibia ESTs
Database: Expressed Sequence Tags (est)
Organism: Amphibia (Taxid: 8292)

Also include the output of that BLAST search in your document. If appropriate, change the font to Courier size 10 so that the results are displayed neatly. You can also screen capture a BLAST output (e.g. alt print screen on a PC or on a MAC press ⌘-shift-4. The pointer becomes a bulls eye. Select the area you wish to capture and release. The image is saved as a file called Screen Shot [].png in your Desktop directory). It is not necessary to print out all of the blast results if there are many pages.



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National Center for Biotechnology Information

Log in

BLAST® » tblastn

HomeRecent ResultsSaved StrategiesHelp

blastnblastptblastxtblastntblastx

Translated BLAST: tblastn

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) ? Clear

P02753

Query subrange ?

From

To

Or, upload file

Choose FileNo file chosen ?

Job Title

P02753:RecName: Full=Retinol-binding protein...

Enter a descriptive title for your BLAST search ?

☐ Align two or more sequences ?

Choose Search Set

Database

Expressed sequence tags (est) ?

Organism

Amphibia (taxid:8292) ☐ exclude [Add Organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown ?

Exclude

☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to

☐ Sequences from type material

Entrez® Query

[YouTube](#) [Create custom database](#)

Enter an Entrez query to limit search ?

BLAST

☐ Search database est using Tblastn (search translated nucleotide databases using a protein query)
☐ Show results in a new window

Feedback

On the BLAST results, clearly indicate a match that represents a protein sequence, encoded from some DNA sequence, that is homologous to your query protein. I need to be able to inspect the pairwise alignment you have selected, including the E value and score. It should be labeled a "genomic clone" or "mRNA sequence", etc. - but include no functional annotation.

Chosen Match: (Result 4) Accession CN058168.1, a 785 base-pair cDNA,mRNA sequence from *Ambystoma tigrinum*. See below for alignment details. Other details include:

Query coverage = 90%

E-value = 4e-93

% identity = 70%

Accession length = 785

i Your search is limited to records include: **Amphibia (taxid:8292)**

Job Title	P02753:RecName: Full=Retinol-binding protein...
RID	RUSP8CKS014 Search expires on 02-01 09:02 am Download All ▼
Program	TBLASTN ? Citation ▼
Database	est See details ▼
Query ID	P02753.3
Description	RecName: Full=Retinol-binding protein 4; AltName: Full=Pl...
Molecule type	amino acid
Query Length	201
Other reports	?

Filter Results

Organism only top 20 will appear ☐ exclude

[+ Add organism](#)

Percent Identity

to

E value

to

Query Coverage

to

[Filter](#) [Reset](#)

- Descriptions
- Graphic Summary
- Alignments
- Taxonomy

Sequences producing significant alignments

Download ▼ Select columns ▼ Show 100 ▼ [?](#)

☒ select all 100 sequences selected

[GenBank](#) [Graphics](#)

	Description ▼	Scientific Name ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
☒	Ag2_p47_I4_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	2e-93	69.61%	731	CN064263.1
☒	Salamander_10_I03.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	3e-93	70.00%	735	CN052017.1
☒	Salamander_20_N22.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	3e-93	70.00%	760	CN055612.1
☒	Salamander_2_G02.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	4e-93	70.00%	785	CN058168.1
☒	Ag2_p35_J10_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	2e-92	70.00%	913	CN062105.1
☒	Ag2_p36_A19_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	3e-92	70.00%	958	CN062231.1
☒	v3_p1_a3_t3 V3 Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	272	272	88%	5e-92	70.62%	697	CN049128.1
☒	Ag2_p38_K1_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	6e-92	70.00%	1012	CN062855.1
☒	Ag2_p38_2_D23_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	1e-91	70.00%	1094	CN062829.1

Alignment Details:

[Download](#) [GenBank](#) [Graphics](#) [Next](#) [Previous](#) [Descriptions](#)

Salamander_2_G02.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence

Sequence ID: [CN058168.1](#) Length: 785 Number of Matches: 1

Range 1: 104 to 643 [GenBank](#) [Graphics](#)

[Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps	Frame
276 bits(706)	4e-93	Compositional matrix adjust.	126/180(70%)	153/180(85%)	0/180(0%)	+2
Query 14		GSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFL				
		S AERDC VS+F V +NFD+ R++GTWYAMAKKDPEGLFL				
Sbjct 104		ASCLAERDCTVSNFPVMQNFDRYRYTGTWYAMAKKDPEGLFL				
Query 74		TAKGRVRLNNDVDCADMVGTFDTEDPAKFKMKYWGVASFL				
		TAKGRV + + +CADMVG FTDT+DPAKF M+Y G+ ++L				
Sbjct 284		TAKGRVVIFGGFQLCADMVGIFTDTDDPAKFVMQYRGLLAYL				
Query 134		VQYSCRLNLDGTCADSYSFVFSRDPNGLPPEAQKIVRQRQE				
		+ YSCR LN DGTCADSYSFVFSRDPNGL PEAQ+IVR+RQ+				
Sbjct 464		LHYSCRRLNDDGTCADSYSFVFSRDPNGLTPEAQKIVRKRQK				

Alignment summary:

Score: 276 bits (706)

Expect: 4e-93

Method: Compositional matrix adjust.

Identities: 126/180 (70%)

Positives: 153/180 (85%)

Gaps: 0/180 (0%)

Frame: +2

In general, [Q2] is the most difficult for students because it requires you to have a “feel” for how to interpret BLAST results. You need to distinguish between a perfect match to your query (i.e. a sequence that is not “novel”), a near match (something that might be “novel”, depending on the results of [Q4]), and a non-homologous result. If you are having trouble finding a novel gene try restricting your search to an organism that is poorly annotated.

[Q3] Gather information about this “novel” protein. At a minimum, show me the protein sequence of the “novel” protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don’t forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don’t have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

Chosen sequence (from EMBOSS TRANSEQ)-

>104-643_1 Salamander_2_G02.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence

ASCLAERDCTVSNFPVMQNFDTRYRTGTWYAMAKKDPEGLFLLDNIVAKFTVDDYGRMLATAKGRVVIFFGGFQL
CADMVGIFTDTDDPAKFVMQYRGLLAYLENGQDPHWVVDTDYDTYALHYSCRRLNDDGTCADSYSFVFSRDPNG
LTPEAQRIVRKRQKELCLDREYRMVVHNGYCE

Here, tell me the name of the novel protein, and the species from which it derives. It is very unlikely (but still definitely possible) that you will find a novel gene from an organism such as *S. cerevisiae*, human or mouse, because those genomes have already been thoroughly annotated. It is more likely that you will discover a new gene in a genome that is currently being sequenced, such as bacteria or plants or protozoa.

Name: Ambystoma novel protein

Species: *Ambystoma tigrinum*

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
Amphibia; Batrachia; Caudata; Salamandroidea; Ambystomatidae;
Ambystoma.

[Q4] Prove that this gene, and its corresponding protein, are novel. For the purposes of this project, “novel” is defined as follows. Take the protein sequence (your answer to [Q3]), and use it as a query in a blastp search of the nr database at NCBI.

- If there is a match with 100% amino acid identity to a protein in the database, from the same species, then your protein is NOT novel (even if the match is to a protein with a name such as “unknown”). Someone has already found and annotated this sequence, and assigned it an accession number.
- If the top match reported has less than 100% identity, then it is likely that your protein is novel, and you have succeeded.
- If there is a match with 100% identity, but to a different species than the one you started with, then you have likely succeeded in finding a novel gene.

- If there are no database matches to the original query from [Q1], this indicates that you have partially succeeded: yes, you may have found a new gene, but no, it is not actually homologous to the original query. You should probably start over.

Details:

A BLASTP search against the NR database (see setup in first screenshot below) yielded a top hit result is to a protein from *Ambystoma mexicanum* (Axolotl). Even though it has a 100% identity, it is from a different species than the one we started with (*Ambystoma mexicanum* vs *Ambystoma tigrinum*)

Other details include 100% coverage and an E-value of 7e-132.

See additional screenshots below for top hits and selected alignment details.

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blastn **blastp** blastx tblastn tblastx
Standard Protein BLAST

BLASTP programs search protein databases using a protein query. more...
Reset page Bookmark

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

TAKGRVVFGGFQLCADMVGIFTDDPAKFVMQYRGLLAYLENGQDPHWVVD
TDYDTYA
LHYSCRRLLNDGTCADSYSFVFSRDPNGLTPEAQRIVRKQKELCLDREYRMV
VHNGYCE

Query subrange [?](#)

From

To

Or, upload file No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ?

Organism [Optional](#)

☐ exclude

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Exclude [Optional](#)

☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ Quick BLASTP (Accelerated protein-protein BLAST)

☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

Search database nr using Blastp (protein-protein BLAST)

☐ Show results in a new window

Feedback

← Edit Search

Save Search

Search Summary ▾

How to read this report?

BLAST Help Videos

Back to Traditional Results Page

Job Title104-643_1 Salamander_2_G02.ab1 AG Ambystoma...

RIDRV7J166R014 Search expires on 02-01 13:16 pm Download All ▾

ProgramBLASTP Citation ▾

Databasenr See details ▾

Query IDlcllQuery_4859546

Description104-643_1 Salamander_2_G02.ab1 AG Ambystoma tigrinu ...

Molecule typeamino acid

Query Length180

Other reportsDistance tree of results Multiple alignment MSA viewer ?

Filter Results

Organism only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity

E value

Query Coverage

to

to

to

Filter

Reset

Compare these results against the new Clustered nr database ? BLAST

Descriptions

Graphic Summary

Alignments

Taxonomy

Sequences producing significant alignments

Download ▾ Select columns ▾ Show 100 ▾ ?

☒ select all 100 sequences selected

GenPept

Graphics

Distance tree of results

Multiple alignment

MSA Viewer

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	retinol-binding protein 4 [Ambystoma mexicanum]	Ambystoma mexicanum	378	378	100%	7e-132	100.00%	200	XP_069467453.1
☒	retinol-binding protein 4 [Ascaphus truei]	Ascaphus truei	298	298	99%	3e-100	77.09%	195	XP_075468936.1
☒	retinol-binding protein 4 [Pleurodeles waltl]	Pleurodeles waltl	295	295	99%	6e-99	75.42%	199	XP_069095936.1
☒	retinol-binding protein 4 [Lissotriton helveticus]	Lissotriton helveticus	294	294	99%	2e-98	76.54%	200	XP_078499520.1
☒	retinol-binding protein 4 [Geotrypetes seraphini]	Geotrypetes seraphini	291	291	99%	2e-97	73.74%	199	XP_033799662.1
☒	retinol-binding protein 4 [Rhinatremas bivittatum]	Rhinatrema bivittatum	290	290	99%	8e-97	73.18%	201	XP_029465754.1

See alignment details here:

Download ▾ GenPept Graphics

Next Previous Descriptions

retinol-binding protein 4 [Ambystoma mexicanum]

Sequence ID: XP_069467453.1 Length: 200 Number of Matches: 1

See 1 more title(s) ▾ See all Identical Proteins(IPG)

Range 1: 16 to 195 GenPept Graphics

Next Match Previous Match

Score	Expect	Method	Identities	Positives	Gaps
378 bits(971)	7e-132	Compositional matrix adjust.	180/180(100%)	180/180(100%)	0/180(0%)
Query 1	ASCLAERDCTVSNFPVMQNFDRYRYGTWYAMAKKDPEGLFLLDNIVAKFTVDDYGRMLA				60
Sbjct 16	ASCLAERDCTVSNFPVMQNFDRYRYGTWYAMAKKDPEGLFLLDNIVAKFTVDDYGRMLA				75
Query 61	TAKGRVVFIFGGFQLCADMVGIFTDDPAKFVMQYRGLLAYLENGQDPHWVVDYDYTYA				120
Sbjct 76	TAKGRVVFIFGGFQLCADMVGIFTDDPAKFVMQYRGLLAYLENGQDPHWVVDYDYTYA				135
Query 121	LHYSCRRLLNDDGTCADSYSFVFSRDPNGLTPEAQRIVRKRQKELCLDREYRMVVHNGYCE				180
Sbjct 136	LHYSCRRLLNDDGTCADSYSFVFSRDPNGLTPEAQRIVRKRQKELCLDREYRMVVHNGYCE				195

Related Information

Genome Data Viewer - aligned genomic context

Identical Proteins - Identical proteins to XP_069467453.1